



EG1113 - C Programming for Engineering Lab

Report #6

Wrapper Classes

Date of the experiment: October 2, 2024

Due date of report: October 4, 2021

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1. Core concepts learned

- Setup a wrapper function for a C program

2. List of equipment and software used

- Raspberry PI 400
- VS Code
- Lenovo Laptop
- Caja File Network Share

3. Relevant web links and other documents:

- Utilized [Raylib](#) to create visual graphics with the PI and to collect input from key presses

4. Exercise Fix & Modify Starter Code

4.1 Summary

- Fix the starter code to run correctly
- Modify the starter code to display a button board on your computer screen

4.2 Procedures

1. Fix the starter code to run correctly by switching defined to undefined and include the initializing steps from part 5

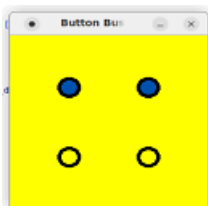
```
if(not_inited)
{
    set_modelpl_400,L_FWD_IN,PI_INPUT); set_modelpl_400,L_BK_IN, PI_INPUT); set_modelpl_400, R_FWD_IN,PI_INPUT); set_modelpl_400, R_BK_IN, PI_INPUT);
    set_pull_up_down(pl_400,L_FWD_IN,PI_PUD_UP);set_pull_up_down(pl_400,L_BK_IN,PI_PUD_UP);set_pull_up_down(pl_400,R_FWD_IN,PI_PUD_UP);set_pull_up_down(pl_400,R_BK_IN,PI_PUD_UP);
    set_modelpl_car,L_FWD,PI_OUTPUT);set_modelpl_car,L_BK,PI_OUTPUT);set_modelpl_car,R_FWD,PI_OUTPUT);set_modelpl_car,R_BK,PI_OUTPUT);
}
```

2. Modify the code to display a button board on your computer screen

```
void draw(int buttons)
{
    BeginDrawing();
        ClearBackground(YELLOW);

        DrawCircle(75, 75, 15, BLACK);
        DrawCircle(250 - 75, 75, 15, BLACK);
        DrawCircle(75, 250 - 75, 15, BLACK);
        DrawCircle(250 - 75, 250 - 75, 15, BLACK);

        DrawCircle(75, 75, 10, buttons & 8 ? BLUE : DARKBLUE);
        DrawCircle(250 - 75, 75, 10, buttons & 1 ? BLUE : DARKBLUE);
        DrawCircle(75, 250 - 75, 10, buttons & 4 ? GOLD : YELLOW);
        DrawCircle(250 - 75, 250 - 75, 10, buttons & 2 ? GOLD : YELLOW);
    EndDrawing();
}
```



5. Exercise Create a wrapper class for Raylib

5.1 Summary

- Modify the code to utilize a wrapper
- Question:
 - If the Raylib graphics library is no longer supported on the Raspberry Pi, the program would fail to compile and run. To prevent lab6.c from such changes, wrapper functions can be created, allowing for easy replacement with alternative libraries while keeping the main application code intact

5.2 Procedures

1. Remove `#include <raylib.h>` from lab6.c
2. Implement a wrapper in `graphics_wrapper.h` and `graphics_wrapper.c` for RayLib

```

C: graphics_wrapper.h <KEY_LEFT>
1 #include <raylib.h>
2 #include "gpio_lib_wrapper.h"
3
4 #define WINDOW_WIDTH 250
5 #define WINDOW_HEIGHT 250
6
7 #define KEY_LEFT_FORWARD KEY_D
8 #define KEY_RIGHT_FORWARD KEY_W
9 #define KEY_LEFT_BACK KEY_A
10 #define KEY_RIGHT_BACK KEY_S
11 #define KEY_RECORD KEY_E
12 #define KEY_LIST KEY_I
13
14 #define KEY_SPEED_MIN KEY_O
15 #define KEY_SPEED_MAX KEY_N
16
17 void draw_buttons();
18 void input();
19
C: graphics_wrapper.c <main>
20 void input()
21 {
22     for (int i = 0; i < buttons; i++)
23     {
24         if (isKeyDown(KEY_LEFT_FORWARD) || isKeyDown(KEY_LEFT_BACK) || isKeyDown(KEY_RIGHT_FORWARD) || isKeyDown(KEY_RIGHT_BACK))
25         {
26             current_speed = speed;
27             break;
28         }
29     }
30
31     buttons = (isKeyDown(KEY_LEFT_FORWARD) || isKeyDown(KEY_LEFT_BACK) || isKeyDown(KEY_RIGHT_FORWARD) || isKeyDown(KEY_RIGHT_BACK)) ? 4 : 0;
32
33     set_pwm(buttons, current_speed);
34
35     if (isKeyDown(KEY_RECORD) && !recording)
36     {
37         saved_i = 0;
38         recording = 1;
39         printf("Recording started\n");
40     }
41
42     if (recording)
43     {
44         saved_value[saved_i++] = buttons;
45         if (saved_i == sizeof(saved_value) / sizeof(saved_value[0]))
46         {
47             recording = 0;
48             printf("Recording is over\n");
49         }
50     }
51
52     if (isKeyDown(KEY_LIST))
53     {
54         printf("Currently recorded:\n");
55         for (int i = 0; i < saved_i; i++)
56             printf("%3d : %2d\n", i, saved_value[i]);
57     }
58
59     draw_buttons();
60 }
  
```

3. Compile and run the code

6. Daydreaming

- Convert the button input to be RGB LED to be certain colors to represent which direction they are going or to represent if they are on or off
- To have the on-screen display to also have trim levels displayed, lights on virtual button board also can be affected by trim levels
- Have a on screen speed display based on how long the buttons are held and then have the lights turn off once the vehicle comes to a stop

7. Problems experienced and areas of concern

- No noteworthy problems

8. Course goals

Rank the level (0=none to 5=a large amount) that this lab covered the each of the course goals by updating the appropriate box in the table below.

Course Goals	Laboratory												
	01	02	03	04	05	06	07	08	09	10	11	12	13
To compile, link, debug and run C programs	4	5	5	5	5	5							
To use the basic C program structure; variables, constants, and operators	4	4	4	4	4	5							
To use the repetition constructs such as looping with for, while, and do-while statements	2	3	4	4	4	4							
To use the selection structures such as if, if/else, switch, conditional expression statement	0	0	3	3	3	3							
To create program modules using functions (passing data to and returning values from functions)	0	0	0	2	4	4							
To use arrays and pointers	0	0	0	0	4	3							
To work with strings and string manipulating functions	2	4	4	1	2	1							
To perform file, I/O operations	3	4	4	5	5	5							
To develop structured, modular, and top-down design of software	0	0	0	1	1	4							